

B.TECH | BCSE497J — PROJECT I

ARIA

Autonomous Reasoning & Interaction Agent

A cognitive, multi-agent vision-language-action software stack for adaptive manipulation planning

TEAM MEMBER

Garv Arora (23BCT0005)

FACULTY GUIDE

Prof. Yuvaraj N

Approval Mail From Guide



Garv Arora 23BCT0005 <garv.arora2023@vitstudent.ac.in>

Project Review 2 – Documents and Approval Request

Yuvaraj N <yuvaraj.n@vit.ac.in>
To: Garv Arora 23BCT0005 <garv.arora2023@vitstudent.ac.in>

Wed, Sep 2, 2026 at 2:54 PM

Dear Garv Arora,

Based on your performance and attached documents (**PowerPoint Presentation** and **Project Review report**), I am providing my approval in the following aspects:

- **Meeting with Guide Frequently:** Satisfied
- **PowerPoint Presentation:** Approved
- **Project Demonstration:** Approved
- **Project Report Draft:** Satisfied
- **Project Outcome:** Ongoing

All the best !!!

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With Regards

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Aim

To design and simulate ARIA — a locally-run, multi-agent vision-language-action system that lets an affordable 5-DoF robotic arm understand natural-language commands, perceive objects using only a monocular camera, plan affordance-aware grasps, and execute reliable pick-and-place manipulation, while continuously monitoring its own confidence, safety, and performance.

Abstract

- ARIA is a locally-run, multi-agent manipulation system built around an affordable 5-DoF arm. It understands spoken or typed instructions, works through them step by step, and picks up, moves, and manipulates objects roughly the way a person would — adjusting for uncertainty, checking in with the user when unsure, and steadily improving its own skills over time.
- Perception is deliberately hybrid: YOLO detection combines with monocular depth (Depth-Anything / MiDaS) and camera-table calibration geometry to recover a reliable Z-coordinate, with no dedicated depth sensor anywhere in the setup. Grasp candidates from GraspNet / AnyGrasp are filtered by an Affordance Agent, and a local LLM Planning Agent orchestrates roughly fifteen specialist agents, pausing for user approval whenever its confidence drops below threshold.

Research Domain Overview

Executive overview of the ARIA Research Literature Compendium — a curated bibliography of 136 papers (2011–2026) underpinning the project's design choices.

136

Total Documents

126

Full PDF Papers

10

Reference Citations

9

Research Domains

#	Research Domain	Count	Key Focus Areas
1	Teleoperation, Demonstration & Robot Learning	10	LfD surveys, GELLO, confidence-aware human-in-the-loop
2	3D Vision, Gaussian Splatting & Spatial Representations	14	3D Gaussian Splatting, spatial scene graphs, 3D VSG
3	Kinematics, Dynamics & Motion Planning	19	AI-enhanced kinematics, multi-robot collaboration, cuRobo
4	Robotics Frameworks, Calibration & Sensor Interfaces	13	ARC-Calib, ROS-Gazebo, MoveIt!/ROS topics
5	Object Pose Estimation, Tracking & Perception	14	Autonomous tracking, EXOT, Speed-FairMOT
6	Vision-Language-Action (VLA) & Foundation Models	20	Embodied CoT, CoT-VLA, Octo, π 0.5, OpenVLA
7	LLM Task Planning, Multi-Agent Reasoning & Decision Making	19	Ambiguity detection, clarification dialogue, SMART-LLM
8	Robotic Grasping & Dexterous Manipulation	21	OOD grasping, stress-minimizing grasp, AffordGrasp
9	Predictive Maintenance, System Reliability & Diagnostics	6	Prognostics, RUL prediction, servo-motor health

Literature Review

Drawn from a curated compendium of 136 papers across 9 research domains (2011–2026): VLA & foundation models, LLM task planning, grasping & dexterity, pose/tracking, kinematics & motion planning, 3D vision, teleoperation, frameworks/calibration, and predictive maintenance.

Paper	Objective	Methodology	Pros / Cons	Findings
π 0.5 (Physical Intelligence, arXiv:2504.16054, 2025)	Open-world VLA generalization	Vision-language pretraining + flow-matching action chunking	+ Strong generalization – Heavy compute	Baseline for evaluating ARIA's VLA skill layer
SMART-LLM (arXiv:2309.10062, 2023)	Multi-robot task planning via LLMs	Decomposes instructions into multi-agent subtasks and allocation	+ Clear subgoal structure – No affordance grounding	Basis for ARIA's Planning Agent hierarchy
Depth Anything V2 (Yang et al., arXiv:2406.09414, 2024)	Monocular depth foundation model	Large-scale synthetic + real depth pretraining	+ Sensor-free depth – Scale ambiguity	Selected for Z-recovery, benchmarked vs MiDaS v3.1
AffordGrasp (arXiv:2503.00778, 2025)	Affordance-driven grasping in clutter	Affordance heatmaps + semantic grasping priors	+ Use-case aware – Needs clean segmentation	Directly informs ARIA's Affordance Agent
SAM 2 (Ravi et al., Meta FAIR, arXiv:2408.00714, 2024)	Instance segmentation in images/video	Promptable segmentation transformer with streaming memory	+ Pixel-accurate masks – Real-time cost	Used for grasp-region accuracy over boxes

Note: limited to 2 slides' worth of coverage, per review guidelines. Full 136-paper compendium available on request.

Literature Review — Gaps & How ARIA Differs (1/2)

Paper	Gap Identified	What ARIA Does Differently
π 0.5 (Physical Intelligence, arXiv:2504.16054, 2025)	Needs large-scale training data / compute; no built-in human-in-the-loop safeguard	Modular, lightweight pipeline on a single consumer GPU with confidence-gated user approval instead of pure end-to-end learning
SMART-LLM (arXiv:2309.10062, 2023)	Plans tasks but has no perception, affordance, or safety grounding of its own	LLM Planning Agent is wired directly to Vision, Affordance, Reachability, and Safety agents via function-calling
Depth Anything V2 (Yang et al., arXiv:2406.09414, 2024)	Monocular depth is relative, not metric — unusable alone for real-world pick coordinates	Combined with camera-table calibration geometry to recover a validated, metric Z-coordinate with no depth sensor
AffordGrasp (arXiv:2503.00778, 2025)	Affordance reasoning evaluated in isolation; no fallback when the model is uncertain	Affordance Agent output is coupled to a confidence-gated approval step before any uncertain grasp executes
SAM 2 (Ravi et al., Meta FAIR, arXiv:2408.00714, 2024)	Produces masks only — no downstream grasp planning or use-case reasoning	Segmentation feeds straight into affordance-based grasp-region ranking, not just bounding-box grasping

Literature Review — Gaps & How ARIA Differs (2/2)

Paper	Gap Identified	What ARIA Does Differently
GraspNet-1Billion (Fang et al., CVPR 2020, arXiv:1912.13470)	Ranks grasps by reachability/geometry only — no notion of how an object is actually used	Affordance Agent re-ranks candidates by use-case (a cup by its handle, a tool by its shaft), not the nearest surface
cuRobo (Sundaralingam et al., NVIDIA, ICRA 2024, arXiv:2310.17274)	General-purpose collision-free planner; not tuned for a hobby-servo arm's trailing cables	Motion planning is explicitly cable-aware and self-calibrating for low-cost SG90/MG995 hardware
FoundationPose (Wen et al., NVIDIA, CVPR 2024, arXiv:2312.08344)	Strong 6D pose but compute-heavy and reliant on CAD/reference models for novel objects	Runs a lighter X/Y/Z pipeline now, with 6D pose kept as a scoped future addition rather than a hard dependency
OpenVLA (Kim et al., arXiv:2406.09246, 2024) / Octo (arXiv:2405.12213, 2024)	Generalist policies act as a black box — decisions aren't easily auditable	Every action carries a confidence score and an explicit chain-of-thought log, so each decision is traceable
SayCan — "Do As I Can, Not As I Say" (Ahn et al., Google Research, arXiv:2204.01691, 2022)	Grounds language in affordances but assumes existing robot infrastructure and no safety loop	Full stack — Safety Agent, Health Monitor, and confidence gating — built around \$150–200 hobby hardware

Research Gap

- Most published manipulation stacks assume a dedicated RGB-D or LiDAR sensor; few recover a reliable Z-coordinate from a plain \$10–30 webcam using monocular depth combined with calibration geometry alone.
- Multi-agent robotics research rarely attempts an agent hierarchy at ARIA's scale — around fifteen cooperating agents — running together on a single consumer GPU paired with sub-\$200 hobby-servo hardware.
- Affordance-aware grasp selection and confidence-gated human approval are typically studied in isolation; ARIA couples the two into one live decision loop.
- Explainable, chain-of-thought decision logging alongside cable-aware, self-calibrating motion planning is uncommon at this price point, limiting transparency in low-cost manipulation research.

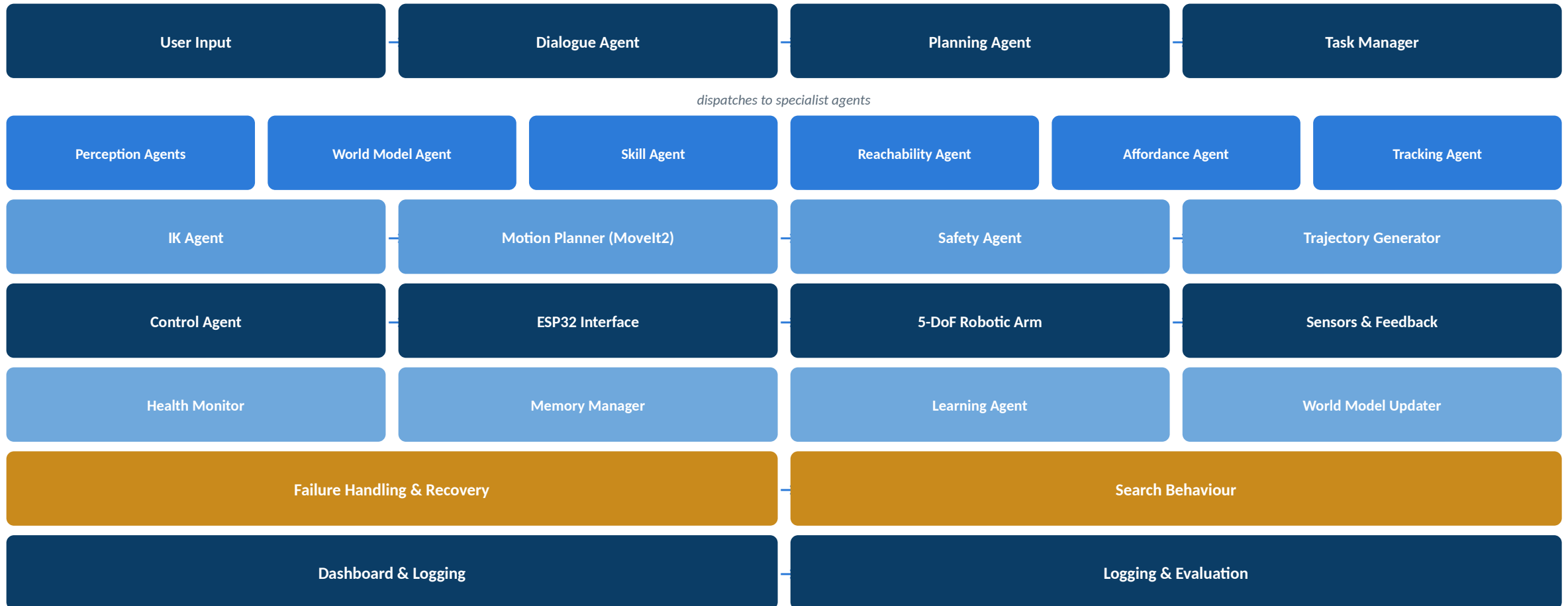
Objectives

- Model and simulate a 5-DoF SG90/MG995 arm with a validated YOLO-based perception baseline.
- Recover reliable Z-coordinates via monocular depth and calibration geometry, without any dedicated depth sensor.
- Build an Affordance Agent that filters GraspNet / AnyGrasp candidates by real-world use-case rather than nearest surface.
- Benchmark inverse-kinematics solvers and integrate MoveIt2 for collision-free trajectory execution.
- Deploy a local LLM Planning Agent that orchestrates specialist agents through function-calling.
- Implement confidence-gated decision-making, a semantic World Model, and self-supervised skill improvement.

Identified SDGs: SDG 9 (Industry, Innovation & Infrastructure), SDG 4 (Quality Education) | **Target Outcome:** Conference / Scopus Publication

Framework / Architecture / Block Diagram

Traces a single command from the moment it's spoken through to the arm moving, then back through the monitoring and learning loop.



Feedback loop: Health, Memory, Learning, and Evaluation data feed back into the Planning Agent to refine future decisions and confidence thresholds.

Functional Requirements

FR1

The system shall accept spoken or typed natural-language commands and decompose them into goals, subgoals, and actions.

FR2

The system shall detect, segment, and localize objects in the workspace using only a monocular camera feed.

FR3

The system shall recover a real-world Z-coordinate without any dedicated depth sensor.

FR4

The system shall generate and rank grasp candidates by how an object is actually used, not merely the nearest reachable surface.

FR5

The system shall plan and execute collision-free, cable-aware trajectories to pick, place, stack, and sort objects.

FR6

The system shall compute a confidence score per action and pause for user approval when confidence falls below a defined threshold.

FR7

The system shall maintain a semantic World Model of known objects, including position, attributes, and lifecycle state.

FR8

The system shall log performance metrics and classify failures to support continuous, self-supervised skill improvement.

Modules

ARIA is organized as ~15 cooperating agent-modules, grouped by function:

Perception

Vision • Depth • Tracking Agents

Object detection, monocular depth, and real-time tracking of moving objects

Cognition & Planning

Dialogue • Planning • Attention Agents

Language understanding, task decomposition, and directing the camera to better views

Manipulation

Skill • Reachability • Affordance • IK / Motion Agents

Grasp selection, workspace-feasibility checks, and collision-free trajectory execution

Safety & Memory

Safety • Memory • World Model Agents

Collision/cable-avoidance, object history, and the running semantic map

Control

Control Agent

Converts motion plans into low-level servo commands via the ESP32 interface

Improvement

Learning • Evaluation • Health Monitor

Self-supervised skill improvement, failure classification, and system health tracking

Experiments and Results

Project is under active development — results below reflect simulation validation completed to date, not final performance.

Completed So Far

- 5-DoF arm and workspace modeled in simulation; YOLO detection baseline validated against spawned test objects
- Z-coordinate recovery via camera-table calibration geometry validated against simulator ground-truth poses
- Depth-Anything and MiDaS benchmarked under varied simulated lighting for the perception pipeline
- IK solvers (ikpy, roboticstoolbox, trac-ik, KDL, analytical) benchmarked on the arm's geometry

Planned Metrics

Pick success rate

vs. baseline over repeated trials

IK / localization error

accuracy of solved joint targets

Planning & inference latency

end-to-end command-to-action time

Failure-category breakdown

missed object, slip, IK, collision, perception

Next milestone: end-to-end pick/place/stack/sort demonstration across a fixed set of simulated test scenes, with results classified by the Evaluation Agent.

Conclusion

- ARIA demonstrates that a carefully wired multi-agent architecture can bring research-grade manipulation behavior to a sub-\$200 hobby-servo arm.
- Hybrid perception (YOLO + monocular depth + calibration geometry) removes the need for a dedicated depth sensor without sacrificing localization accuracy.
- Confidence-gated, affordance-aware grasping and an explicit chain-of-thought log give the system both safer behavior and end-to-end explainability.
- Simulation work completed so far — perception baseline, Z-recovery, and IK benchmarking — validates the core pipeline ahead of full agent integration.
- Remaining work focuses on integrating the Planning, Learning, and Evaluation agents end-to-end and demonstrating pick/place/stack/sort tasks with logged, reproducible metrics.

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